

Swamps along the Manawatu and Oroua Rivers were very large but not all were clearly delineated on the early plans. The Makerua swamp (or Makurerua swamp on the early plans) filled a depression between the Manawatu River and the low terraces running between Linton and Tokomaru. In the lower reaches of the Manawatu River a similar swamp occupied the low ground now traversed by the floodway. Several large swamps bordered the west bank of the Oroua River between Rangiotu and Feilding. A small swamp filled a depression at the south end of Poutu Rd near the Rangitikei River. Swamps were extensive in the Taonui basin. The Taonui swamp lay near the confluence of the Manawatu and Oroua Rivers, and the Kowhai and Te Waiti swamps a little further to the north. Between these and the Oroua River J. T. Stewart (S.O. 10603) wrote "a swamp called Te Pukakurau is about here".

The plans show swamp vegetation associated with the coastal lagoons and comments indicate that there were swampy flats throughout the sand country and in numerous places on the margins of the rivers.

Shrubland (or scrub) is applied broadly to vegetation dominated by low shrubby species and is frequently used to denote vegetation varying from grassland with prominent shrubs to short open forest. Little reference is made to this type of vegetation in the survey plans. A few patches are marked along the Manawatu River. J. T. Stewart in his Awahou map (S.O. 10639) coloured "small bush and scrub" light green but this does not show on the monochrome prints available to me.

THE FORESTS

Manawatu forests fell broadly into the following divisions:

- (1) Semi-swamp forest dominated by kahikatea and pukatea in the low-lying land near the rivers.
- (2) Totara forest on the free-draining soils and the low-rainfall areas.
- (3) Mixed podocarp forest composed of rimu, matai, totara, and kahikatea occupying parts of the Manawatu plains, the terraces east of the Manawatu River, and the basins of the Tiritea and Kahuterawa Streams.
- (4) Tawa forest in the greywacke hills, on the terraces between the Manawatu Gorge and the Kahuterawa valley, and on parts of the plains.
- (5) Black beech forest at Aokautere.
- (6) Northern rata-kamahia forest with podocarps on the upper flanks of the Tararua Range.
- (7) Forests of the dunes composed chiefly of tawa, mahoe, and titoki.

Semi-swamp forest

THE PAST

Distribution

Semi-swamp forest dominated by kahikatea and pukatea was a conspicuous feature of the lowlands on both the flood plains and some of the wet sand plains.

This forest was most extensive on the low-lying land fringing the Taonui basin formed within the levées of the Manawatu and the Oroua Rivers near their confluence. Elsewhere there was a mosaic of semi-swamp forest with other forest types along the margins of the rivers and streams. There are relics which show that kahikatea and totara often grew in close proximity. Presumably the totara was mostly on the higher ground on the coarser alluvium while kahikatea grew on the finer sediments which remained wet or very moist throughout the year.

Remnants suggest that kahikatea grew near the western margin of the Makerua swamp on land which was occasionally flooded by the Manawatu River. An excellent example remains at Linton near the railway line. The eastern margin of this swamp abutted low terraces between Linton and Tokomaru and it seems unlikely that there was much more than a ribbon of semi-swamp forest there. It is interesting to note that B. C. Aston (1920b, page 110) described the Esplanade bush in Palmerston North as "virgin forest of tawa, pukatea and white pine". However, this site was probably considerably drier than the typical semi-swamp habitat.

Semi-swamp forest of the lower Manawatu

In the lower parts of the Manawatu River below its junction with the Oroua River, semi-swamp forest tended to occupy higher ground, the concavities being filled with swamp. On the Moutoa Estate, for instance, kahikatea grew on the narrow levées but *Phormium* and raupo filled the large depression (Poole 1946, Poole and Boyce 1949). On the margin of a stretch of the river known as "the Long Reach" early maps (Stewart S.O. 10639) have the comment "small timber". Presumably this was the kahikatea mentioned by Poole and Boyce and which still remains in fragments in that district. It may have been milled to some extent before the first maps of the district had been drawn as E. J. Wakefield (1845, volume 1, page 356) recorded boat-building there in 1840. Further up stream at Poplar Rd J. T. Stewart's map (S.O. 10639) shows "tawa trees", marked across a patch of riparian forest. It is clear from this that other species as well as kahikatea and totara grew near the river. The apparent absence of totara in the lower reaches of the Manawatu may be associated with predominance of fine alluvial

material in that region. Surviving stumps show that totara grew in that district at an earlier period.

Semi-swamp forest of the sand country

The semi-swamp forest on the sand plains probably differed in no essential way from that on the alluvium. Patches of forest developed on the third phase plains and did not occupy all the plains which seem to have offered a suitable habitat. Because of their association with this phase they are mostly well inland, except Round Bush (which was used as a landmark for coastal trading boats) and several other pieces near Foxton. It must be borne in mind that most of these wet plains were considerably wetter before they were drained for farming. However, some of the semi-swamp habitats have proved difficult to drain and still bear the original forest cover more or less intact except for the removal of the timber trees and damage caused by cattle. In these places fairly accurate recording of original vegetation and soils is possible. Where no forest remains we cannot be sure what sort of forest existed as there appear to be no written records. We can be thankful that at least we have fairly accurate forest margins marked on the early plans.

Patterns of soils and primitive vegetation on the sand country do not coincide closely. The forests occupied only some of the sites which appear to have been favourable to them and there are some soils which are fairly uniform throughout but were forested in some places and not in others. This suggests that forests were removed from some parts not very long before European settlement began.

The sandy soils on which semi-swamp forest developed are melanised in the top 200–300 mm and some have a little peat. They are all gleyed and have mottles and usually also concretions in the subsoil. These have been mapped by Cowie and Smith (1958) as Pukepuke and Carnarvon soils. The particular soils on which semi-swamp forest is certain to have developed are Pukepuke brown loamy sand, Pukepuke brown sandy loam (and its peaty phase), Carnarvon brown sandy loam, and Carnarvon brown loamy sand. Forest of some sort also grew on some better-drained sand plains on soils mapped as Pukepuke brown sand but it is unlikely that kahikatea and pukatea were very prominent on them.

Composition and structure

Of the nature of the primitive semi-swamp forest we know little apart from what the meagre early accounts and the vegetation remnants tell us.

L. Cockayne's description of the vegetation of a scenic reserve near Levin (Cockayne 1928, page 176) could have fitted the semi-swamp forests a few

Figure 5 Roots of kahikatea exposed after drainage and soil compaction lowered the land surface on the margin of Buchanans Bush. 1962



miles north in the Manawatu: "Where the ground is wet rise up huge much-buttressed trunks of *Laurelia*, its far-extending roots putting forth abundant pneumatophores. Finally, there are the straight trunks of *Podocarpus dacrydioides* and their crowns full of *Astelia solandri* looking like huge birds' nests".

Aston (1920a, page 63) commented on the state of Round Bush before 1920. It is apparent from his remarks that he saw the reserve prior to the removal of the kahikatea. He wrote, "An excellent example of a swampy forest is the Round Bush at Foxton, where, surrounded by high sandhills, on an area on the sand, tall white-pine forest has developed, although the floor is completely covered with stagnant water. One can walk through the forest dryshod only by treading on the gigantic tree roots which have raised themselves above the soil and water . . .".

Semi-swamp forest at Kopane

In the Kopane district semi-swamp forest and totara forest grew side by side, the totara mostly on the levées. Mr J. Milligan (pers. comm.) moved into the district in 1913 and clearly recalled many of the relics of forest which were cut over for a second time after he settled there. North of the Kopane swamp, and to the north and west of Mill Rd near Sluggish Creek there were heavy stands of kahikatea. A mill near Mr Milligan's house cut "a million feet" (presumably super feet) of kahikatea and the timber was sent to Cook, the cooper, in Palmerston North. Another mill operated for a year near where Rongotea Rd crosses Sluggish Creek, and another by the same creek on Taipo Rd. The latter cut much of the early timber in the district. As some kahikatea was left standing until the second milling the first cut was probably of totara and the superior trees of kahikatea. Mr Milligan commented that the kahikatea on his farm was easily removed as the ground had sunk and the sheep could walk under the roots of the trees (*figure 5*). He mentioned the presence of black maire, white maire, and two forms of tawa distinguished as "black tawa" and "white tawa" on the colour of the wood. There was no kamahi and only occasional rimu. It is possible that rimu and other timber trees had been removed earlier.

Figure 6 Kahikatea in Kitchener Park. The heads of cabbage trees stand above *Muehlenbeckia australis* and New Zealand passion vine. 1962



Figure 7 A relic of kahikatea forest in Buchanans Bush in the Taonui basin now protected as a private reserve. Most of the present trees established after milling. 1962



THE PRESENT

Distribution

Today semi-swamp forest is confined to a few pockets. At Round Bush and Himatangi Bush kahikatea has been felled but enough of the communities remain intact to give an impression of the original vegetation. Groups of misshapen kahikatea grow in Keebles Bush, Kitchener Park (*figure 6*), the Esplanade, and some other places but are so limited in extent that they have little of the character of semi-swamp forest. Along the Manawatu River and at Buchanans Bush in the Taonui basin (*figure 7*) are small stands of pole kahikatea. These have developed since milling, have few associated species and bear little resemblance to the semi-swamp forest that the early settlers saw.

Round Bush

Round Bush has semi-swamp forest in its low-lying parts. The kahikatea were removed apparently about 1918, soon after Aston's visit but there is a magnificent stand of pukatea which, though not tall or stout by some standards, is typical of the species in forest which has not suffered serious damage. There are about 4.7 stems per 100 sq. m up to a metre in diameter—but most are 0.3–0.6 m—reaching a height of nearly 28 m and many are festooned with kiekie to about 18 m. Nearly all have dead crowns. The clumps of *Collospermum bastatum* (referred to by Cockayne as *Astelia solandri*, the name then used for the plant) are a striking feature. *Griselinia lucida* is also prominent. *Metrosideros perforata*, *M. diffusa*, and kiekie reach high into the canopy on their supporting trees. *Metrosideros fulgens* and *M. colensoi* also occur in semi-swamp forest but do not climb as high (see also Esler and Greenwood 1968).

Himatangi Bush

In Himatangi Bush the damage has been more severe. Where the malformed pukatea and the few unmillable kahikatea do not make a closed canopy several smaller trees and shrubs have risen to prominence. Among them are mahoe, pigeonwood, titoki, mapou, cabbage tree, kaikomako, *Dicksonia squarrosa*, and nikau, often well interlaced with supplejack and *Rubus cissoides* (figure 8). In some places *Coprosma propinqua* and *C. areolata* predominate.

The richness of the understorey varies widely with the wetness of the habitat and the amount of light reaching the forest floor. Wet and dark interiors limit the range of shrubs and small plants while drier and better-lit environments are more favourable. Milling, browsing by cattle, and drainage may have altered the structure and habitat of the forest in such a way that a more varied flora has been promoted.

On the wettest parts of the forest floor there are relatively few small plant species. *Ranunculus rivularis* grows in the streamlets, and *Callitriche muelleri* and *Scirpus inundatus* spread across the wet mud. In places where a greater amount of light reaches the floor three herbs are usually found—*Callitriche stagnalis*, *Myosotis caespitosa*, and *Galium palustre*. Supplejack invariably occurs in these situations and, because of the barrier that it forms, has protected other plants from grazing animals. Thus the mounds standing well above the winter water level support a wide range of plant species which have no dependent relationship with semi-swamp forest. Where there is grazing the mounds are occupied by a few species other than *Oplismenus imbecillus* and *Blechnum filiforme*. Severe damage to the forest is accompanied by a change of floor dominants among which wetland pasture species are prominent, e.g., *Juncus sarophorus*, *J. gregiflorus*, *J.*



Figure 8 Interior of Himatangi Bush. Milling of kahikatea, trampling, and browsing have promoted an abundance of supplejack on the wet flats. *Oplismenus* and *Solanum nodiflorum* grow on the forest floor.

1969

articulatus, *J. effusus*, and creeping buttercup. *Carex secta*, *C. virgata*, and *Gabnia xanthocarpa* occur less frequently. This is the final stage in the transformation of forest to pasture aided by milling, grazing, and drainage. Farmers then push over the few remaining trees with a bulldozer and all resemblance to a semi-swamp forest is lost.

Drier conditions on the forest floor favour a range of species which are just as much denizens of any moist forest as of semi-swamp forest. The pukatea stand at Round Bush has on its more elevated parts an understorey of

Geniostoma, *Coprosma areolata*, kaikomako, *Pteris macilenta*, *Asplenium bulbiferum*, and *A. lucidum*. At Himatangi Bush grazing animals have kept the *Geniostoma* and *Asplenium* spp. in check but there are many additional species not present in Round Bush. The conspicuous ground plants at Himatangi Bush are *Hydrocotyle novae-zelandiae*, *Cardamine debilis*, *Urtica incisa*, *Blechnum capense*, *Clenitis glabella*, *Uncinia uncinata*, *Carex solandri*, *C. dissita*, and *Oplismenus imbecillus*. The orchids *Corybas trilobus* and *Pterostylis banksii* are to be seen in spring and early summer. Although typically not a wetland plant, *Fuchsia perscandens* occurs in several remnants which have some semi-swampy ground. It is plentiful at Himatangi Bush and Bledisloe Park but diminishing at Kitchener Park, its type locality.

Totara forest

THE PAST

Distribution, composition, and structure

Little can be written about the totara forest for which the district was well known because not a vestige remains as it was a century ago.

Long before white men saw the Manawatu large areas of totara forest were destroyed by changing levels of the Manawatu River. Some time later it seems that totara forests may have been destroyed in Maori fires on the downlands. In this region, which receives less than 914 mm of rain a year and is subject to occasional drought and fairly constant drying winds, totara forest existed near the climatic margin of forest and grassland. Fires would have been more damaging and regeneration much slower than in higher rainfall areas.

There is little visible evidence now of the full extent of the early totara forests. Stewart's map (S.O. 10603) indicates that totara was encountered in many places along the Manawatu and Oroua Rivers upstream from their confluence at Puketotara. There may be some significance in the name Puketotara itself but it is unwise to translate to suit a purpose. Moving up the Manawatu River from this point the first place where "totara trees" is marked is at the end of Whitmore Rd and on the other side of the river in the direction of the Longburn Freezing Works. The sites of both are traversed by the main trunk railway.

"Totara and matai" is marked on land now occupied by Linton Military Camp. Near Palmerston North, totara stands are indicated at Terrace End bordering the northern margin of the Papaioea clearing and also across the Manawatu River on the Aokautere flats and near Te Matai Rd, the latter commented upon by Kettle (1842). No mention was made of the Hokowhitu totara forest.

Totara is indicated on both sides of the Oroua River, at the end of Kaimatarau Rd. E. J. Wakefield (1845, volume 2, page 235) described it as a stand of "magnificent totara trees". Totara is shown again in several places near Aorangi.

Because survey lines were run along the rivers the riparian totara stands could not escape notice. However, it is probably safe to assume that few if any totara stands existed on the plains more than one kilometre from the rivers, except for a block between Taonui and Bunnythorpe.

Totara on terrace country received no mention at all on early survey plans. To the east of the Manawatu River to the crest of the range in some places there are relics of totara but this species probably formed a minor part of the forest.

What was probably the largest block of totara forest in the Manawatu grew on the terraces of the Rongotea-Ohakea district. In notes written for me shortly before his death the late Edgar Dear stated: "Take a line from Rongotea cemetery to half way to Taipo Rd and then to Ohakea aerodrome—this was the finest area of totara in the Manawatu". Recalling what remained of the forest in his youth he listed matai, miro, rimu, and northern rata as the associated species of tall trees. He wrote: "Some of the largest rata that ever grew surrounded the present town of Rongotea. These were too large to be felled but were killed by fire when the area was burnt. These trees were a feature of the landscape for 25 years after the bush was felled". He thought that some of the trees were nearly 90 m high but this was probably an over-estimate. The smaller trees he listed for the forests of the area included black maire, white maire, rewarewa, hinau, mahoe, kowhai, ngaio, lancewood, fuchsia, and "five varieties of lacebark".

North of Taipo Rd forest was limited to quite small pockets, mostly in gullies. In the many relics (figure 9) there is a strong suggestion of a predominance of totara in the early days, except on the wetter flats where other podocarps made up a large proportion of the canopy. The regeneration of totara on some roadsides and in the fenced remnants is testimony of the suitability of climate and soil for totara.

The downlands carried manuka, toetoe, koromiko, tutu, and bracken. In places there were tree stumps giving credence to my belief that forest occupied some or most of the scrub country not long before European settlement. It is likely that totara grew on this country also.

THE PRESENT

Distribution

There are isolated totaras in forest remnants and on farmland but these are little more than markers of the sites of former forests of totara. In many



Figure 9 A pocket of totara forest on Ngaio Rd near Sanson. When farming began on the downlands of the Mt Biggs district there were only very small pockets of forest. Most of these have persisted to the present day. 1969

cases the trees existing today are quite young, probably having established about the time the timber trees were removed. The few moderately large specimens are too misshapen to have been of any interest to the sawmillers.

The absence of good remnant forests is easily explained. Large quantities of timber, of totara in particular, were used to furnish the needs of settlers. Poorer trees that were not used at that time were cut out later for fence posts.

Composition and structure

There was little forest on the downlands at the time of settlement, but at present there are a greater number of native species than in any part of the Manawatu formerly in continuous forest. It seems that there was a desire among farmers in this region with little timber, firewood, and shelter to retain the minor areas of forest. More recently their efforts have turned towards active preservation. It is worthy of note that these remnants have fared better than Kitchener Park and Awahuri Scenic Reserve which were set aside as public reserves for the preservation of scenery and flora of the district.

The composition of the totara forests of the early days can be judged only by the species that grow on the downlands now, but it is necessary to bear in mind that many of the present-day abundant species in regenerating forest may have been minor elements in the primary vegetation. Nearly all remnants have at least some totara and matai (and kahikatea in low-lying places) but there is no rimu. The most frequent smaller trees are titoki, tawa, black maire, white maire, *Plagianthus betulinus*, lancewood, *Hoheria sexstylosa*, *Pittosporum eugenoides*, *P. tenuifolium*, pate, and mahoe. There is remarkably little pigeonwood, probably because most of the sites are a little dry for it. The major shrubs include rangiora, *Carmichaelia flagelliformis*, *Melicope simplex*, *Paratrophis microphylla*, *Melicytus micranthus*, kaikomako, and *Teucrium* which is rare elsewhere in the district. Vines are abundant because the remnants are small and damaged: about 17 species are frequently encountered. The ferns occurring in most remnants are *Asplenium bulbiferum*, *A. hookerianum*, *A. flaccidum*, *Hypolepis tenuifolia* (in two forms), *Pellaea falcata*, *P. rotundifolia*, *Phymatodes diversifolium*, *Pyrrosia serpens*, and *Polystichum richardii*. The monocotyledons and the herbaceous dicotyledons are very numerous but few are true forest species. Those which are likely to have been in the primary forest include *Carex dissita*, *C. ochrosaccus*, *C. lambertiana*, *C. forsteri*, *C. testacea*, *Uncinia* spp., *Hydrocotyle* spp., and two species not known to occur elsewhere in the Manawatu—*Senecio rufiglandulosus* and *Australina pusilla*.

This account gives little of the nature of the early totara forest. The preservation of just a few hectares of the original totara forest in good condition would have given us something of immense scientific value. R. M. Greenwood's description (Greenwood 1949) of Totara Reserve in the Pohangina valley must serve as an account of riparian totara forest of the Manawatu.

Totara logs in the lower Manawatu

Some reference needs to be made to the forests containing totara which existed in low-lying areas along the Manawatu River in the distant past. These are marked by abundant logs and stumps in swampy land. In some places the logs have been extracted, split, and used for fence posts. Some still retained the bark and the pink colour of the wood but this soon darkened on exposure to air. The posts were durable when used on swampy land but had a very short life in sand. Possibly some logs reached their resting place in flood waters but local farmers do not accept this suggestion as there is good evidence to show that the trees grew where they were found. Such evidence is provided by patches of stumps with roots *in situ*. They were on the Barber Estate midway between East Base trig and Himatangi Bush and known in the early days as Go-to-Hell Swamp. I visited one of these with Mr J. W.

Barber. At the time of settlement the flat was covered in *Phormium* growing on peat 1.8 m deep. Now, following drainage, the peat has sunk and the jagged stumps protrude above the surface. The stumps are large and spaced 1.8–3.6 m apart, indicating that a dense forest once grew there. As well as totara it was possible to identify matai, rimu, kahikatea, and one species of maire. Some farmers have the impression that a gale may have snapped the forest off above ground level. It seems more likely that a raised water table killed the trees and swamp filled the flat thus preserving the bases of the trees while the projecting tops rotted away. I believe that the outfall of the Manawatu River may have been restricted periodically. This could have been caused by accumulation of sand at the mouth or by earth movements. A small anticline running approximately NNE from Whirokino is evident from the drainage pattern of the sand country, the profile of the warped coastal plain, seismic reflection and gravity survey data (Rich 1959). Evidence for recent movement of folds in the area was proposed by Te Punga (1957).

Totara logs occur in many other places in the coastal region. Behind the fore dune along the full length of the coast totara logs in an advanced state of decay are exposed. These mark the shore line a century or more ago and were water-borne. Mr Barber said that totara logs are encountered at varying depths just north of Foxton up to one kilometre east of State Highway 1 and he is sure they have been deposited by the sea. Near the coast south of the Manawatu River it seems that totara grew on the sand country quite close to the sea.

It is my belief that totara grew on the older dunes in the Manawatu but supporting evidence is very slender. At Whites Bush near Foxton numerous massive totaras grow on dunes of the third dune phase but the species has not been seen in a similar situation elsewhere. The vulnerability of vegetation of dry dunes to fire and the usefulness of totara for timber in a timber-deficient part of the district could account for the general absence.

Podocarp-broadleaf forest

THE PAST

Distribution

This category includes stands dominated by podocarps but is intermediate between semi-swamp forests on the very wet soils and totara on the droughty soils. In many cases, podocarp-broadleaf forest contains kahikatea, pukatea, and totara as minor constituents. Tawa is invariably present. This type of forest occupied recent soils, parts of the terraces east of the Manawatu River, and some flat to undulating land near the heads of the Tiritea and Kahuterawa Streams.



Figure 10 Terraces near Linton. In reference to this region a surveyor in 1868 commented, "Fine level land. Rimu and tawa bush." Ryegrass-white clover pastures now dominate but there is some secondary forest in the gullies. 1971

The extent of forest of this nature east of the Manawatu River will never be known with certainty but numerous minor diminishing remnants suggest that mixed podocarp forest covered much of the terrace country from Scotts Rd to about Pahiatua Track with some local variations. The plan of Fitzherbert (S.O. 10837) drawn by J. Knowles in 1868 is marked "Fine level land. Rimu and tawa bush" near a stream called the Nguturoa which crosses the State highway near Milricks Line (figure 10). A patch of very degraded forest on Pahiatua Track (grid ref. N149/160320) has enough of its larger species remaining to indicate the composition of the original forest. Tawa is now the most conspicuous tree but rimu and kahikatea are well represented. In addition there are matai, miro, totara, pukatea, hinau, black maire, and titoki. There are numerous stumps, apparently of podocarps, but these are not readily identifiable.

Composition and structure

The vegetation of the Manawatu plains is very poorly documented. Stewart's map (S.O. 10603) has inscribed across the plains: "Flat land. Totara, rimu, matai, tawa and other timber". This seems to be the most detailed, and in fact, the only description of the early forest available to me. Interviews with old residents filled some of the gaps. All said that towards the low-lying areas generally south of the Kairanga school kahikatea outnumbered all other trees while in the Bunnythorpe direction totara became more abundant. The nature

of the intervening forest was recalled for me by Mr Keith Thompson who farmed along the Oroua River near Kopane. Near his house the composition of the canopy-forming trees in a forest remnant—tawa, titoki, and a few kahikatea—has not changed greatly in his lifetime but he witnessed the removal of the last of the forest between his farm and Palmerston North. Matai was by far the most important tree with rimu ranking next. Apparently there was very little northern rata, a view confirmed by the late Mr Bruce McEwen.

A glimpse of the mixed podocarp forest of the terraces east of the Manawatu River is afforded by Keebles Bush (*figure 11*) if due allowance is made for the damage caused by the 1936 gale which reduced the number of emergent trees to about half, and the 1969–70 drought (Atkinson and Greenwood 1972). According to Mr C. T. Keeble the gale toppled the kahikatea and broke the rimu. He said there was very little northern rata in the region.

Most of Browns Flat was cleared about the turn of the century, yielding a large quantity of very high quality timber, mostly rimu (*figure 12*). The logs were taken via a tram track down the Kahuterawa valley (*figure 2*) for milling. The track is now used as a farm road. Presumably Kahuterawa Flat was cleared about the same time and rimu was the main species. There has been some more recent felling nearby on the Tararua Range and a few rimu logs are still being extracted along with some northern rata for firewood.

Both flats are littered with logs and stumps. Northern rata predominates because of its durability but there is still evidence showing that black maire was an important associated tree. Logs of this species are abundant, many having been pushed into the water channels to serve as bridges when the trees were felled. The only clearly distinguishable podocarp is rimu. Trees were sawn high above the ground so that only the very best of the trunk was taken for timber. This was the practice in the days when native timber was still plentiful. I have little doubt that matai and miro were also milled and believe that kamahi was an important constituent of the forest though there are no logs there now to confirm this view.

Forest of the same general type formed narrow strips in other places along courses of the Kahuterawa and Tiritea Streams.

THE PRESENT

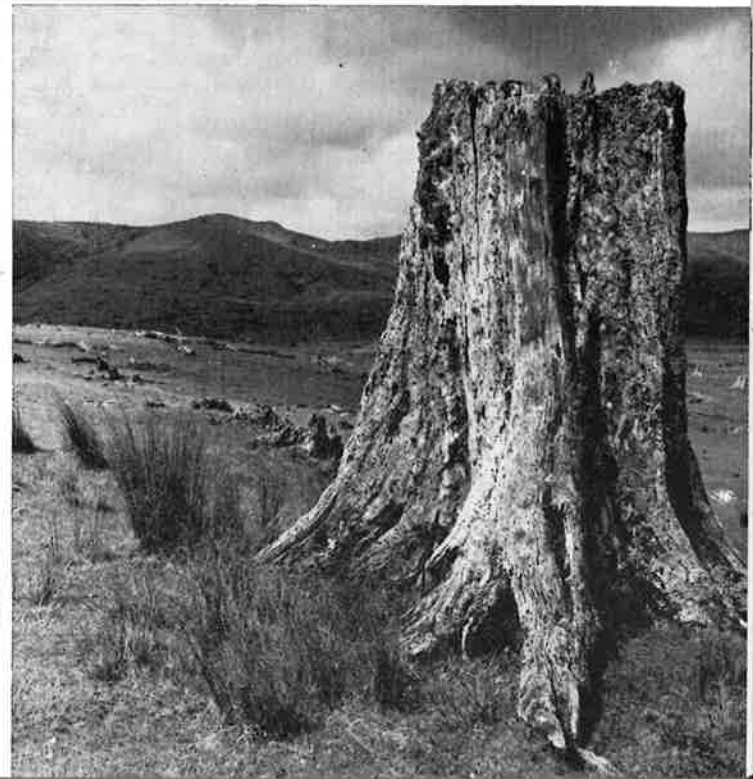
Distribution

In spite of the damage it has suffered Keebles Bush in parts still retains some of the character of the original forest and is the only relic of this type which contains numerous podocarp trees. Kitchener Park has been so grossly modified that it is difficult to judge its original composition. It was probably

Figure 11 Keebles Bush is the finest piece of podocarp-broadleaf forest in the Manawatu. 1962



Figure 12 Browns Flat was forested until about the end of last century. It yielded good quality rimu timber in the days when timber was still fairly plentiful. 1969



podocarp-broadleaf forest. Fragments that remain in other places have had all the podocarps removed and are either very degraded or have become tawa-dominant. These are mostly unfenced patches of bush unlikely to survive much more than a decade or so.

In the relatively undamaged areas of Keebles Bush the podocarps are fairly widely spaced but there are parts where the density is about as high as it would have been before the 1936 gale. Rimu is the most conspicuous tree (figure 13) but there are about the same number of matai stems and fewer kahikatea, totara, and miro. The tallest podocarps far exceed 30 m in height and the diameters range from 30 to 152 cm. These are emergent rather than canopy-forming, the main canopy lying between 9 and 15 m with a few trees reaching above this. The major constituents are tawa (in almost pure stands in places) and mahoe but there are in addition, titoki, kamahi, rewarewa, hinau, *Pittosporum tenuifolium*, mapou, and lancewood. This is one of the last places in the Manawatu where there is still an appreciable amount of kamahi remaining. This is probably due to the low opossum numbers evident from the undamaged lemons, gooseberries, and roses growing nearby. Opossums, almost certainly responsible for the destruction of northern rata in other parts of the district, are free of blame for the absence of northern rata from this reserve as there was very little northern rata in the region in the early days.

Pate, wineberry, *Coprosma australis*, *Geniostoma*, and tree ferns overhang the Maungatungaroa Stream (figure 14) that winds through the reserve and *Dicksonia squarrosa* covers most of the narrow floodable terraces. Kawakawa is the predominant shrub but there are appreciable amounts of *Paratrophis microphylla* and *Geniostoma*. Of lesser importance are *Melicytus micranthus* and the juveniles of kamahi and rewarewa. Podocarp regeneration is negligible except on the steeper aspects where there are some saplings.

Vines in this least damaged part of the reserve are relatively few except for supplejack (which abounds in moist depressions), *Metrosideros diffusa* and New Zealand passion vine. Of the 29 species of ferns only *Asplenium bulbiferum* is abundant. *Phymatodes diversifolium* and *P. scandens* rank next in importance. Occasional patches of *Hymenophyllum demissum* are encountered on the forest floor but the other 4 species of filmy fern are not conspicuous.

The most degraded part has a low discontinuous canopy of mahoe and titoki with an understorey of kawakawa and *Paratrophis microphylla*. The most abundant vines are *Parsonia heterophylla*, *Muehlenbeckia australis*, *M. complexa* var. *grandifolia*, supplejack, *Rubus cissoides*, and New Zealand passion vine. These have replaced most of the trees and shrubs in some places.

Kitchener Park (figure 6) was a landmark since the district was first settled and was admired as a scenic attraction. According to a report in

Figure 13 Rimu in
Keebles Bush. 1962



the *Feilding Star* on 2. January 1916 it had changed little in the 35 years prior to its opening as a public reserve under the control of the Feilding Borough Council in 1916. When Allan (1924c) reported on the vegetation he commented that the reserve was in a much modified condition. At this time it contained 140 native plant species. The number is now about 80. Trees may have been removed from the bush in the very early days but there is no confirmation of this. Just how long farm livestock have had access is uncertain but it is clear that browsing has been the main cause of deterioration in recent years.

The tallest trees, kahikatea and matai, are now few. Totara occurs mostly along the western margin. Tawa, the main canopy species, is probably much more abundant than in the original forest. From the brief description given



Figure 14 The margin of the Maungatungaroa Stream in Keebles Bush is one of the few stream banks in the Manawatu with native vegetation not disturbed by sheep and cattle or human activity. Pate, wineberry, *Coprosma australis*, *Geniostoma*, and tree ferns support kiekie and supplejack. 1962

by Allan it appears that the main shrub layer consisted of small-leaved species together with some tree ferns. Several of the small-leaved species (*Pseudopanax anomalus*, *Melicope simplex*, *Melicytus micranthus*, *Coprosma rotundifolia*, *C. areolata*, and *C. rigida*) are still present but only as scattered individuals. The only shrub present now in quantity is kawakawa. This species was not mentioned by Allan and presumably was then only a minor constituent. *Tradescantia* now forms a blanket over most of the forest floor and is suppressing nearly all regeneration.

Trends in forest degeneration

Most forest remnants in the Manawatu are in a degenerate state and all stages of deterioration are represented. In nearly all instances the chain of events began with the removal of timber trees. Without the entry of cattle into the bush the gaps would have soon filled with broad-leaved species, or possibly kahikatea or totara, depending on local conditions. There was almost

no regeneration of matai, miro, or rimu. Where tawa has been able to form a pure, almost continuous canopy the species has survived while most isolated and exposed trees have died. The effect of cattle has been the removal of palatable species and the encouragement of less palatable small-leaved shrubs.

All remnants provide ample habitats for vines because the margins are relatively extensive and there are many canopy gaps. The native climbers which assume some importance in this situation are *Muehlenbeckia* spp. *Calystegia tuguriorum*, New Zealand passion vine, and *Parsonsia heterophylla*. *Clematis vitalba* is the most significant alien climber.

A feature of remnants at an intermediate stage of degeneration is the remarkably high number of native plant species they contain though the number of individuals of some may be small. This is the state of many of the remnants on the downlands. At this stage there is some chance of rehabilitating a patch of bush for its scenic and biological interests by fencing out livestock. Once *Tradescantia* makes an entry few small plants have any chance of survival and, by suppressing regeneration of the canopy species, this weed can spell doom to a remnant.

At a relatively late stage *Solanum pseudocapsicum* and *S. diflorum* appear and can persist under browsing and trampling conditions which are too severe for most native shrubs to tolerate. When this treatment is accompanied by thinning of the canopy the forest floor becomes grassy. What usually remains is a clump of gnarled mahoe festooned with *Pyrrosia* and *Phymatodes diversifolium* and massive vines of *Rubus cissoides* and *Parsonsia heterophylla*. The remnant then has neither use nor beauty and the final stroke is made with a bulldozer.

Tawa forest

THE PAST

Distribution

Tawa is a component of all forests except beech forest. In some places it assumes dominance. The rise to dominance has been aided by the removal of podocarps which once formed a high canopy above it. This has occurred in two phases. Over a span of many centuries the podocarps failed to produce sufficient numbers of surviving younger members to ensure a prominent place in the vegetation. Within the last century the swing towards tawa dominance has been hastened by the felling of podocarps for timber and posts. Thus a number of remnants such as Kitchener Park almost certainly have more tawa than before milling.

The extent and distribution of tawa forest at the middle of last century is not known in detail. Stewart's maps (S.O. 10603 and S.O. 10639) identified